CS310 NLP Assignment 2 Written Solution

12311805 莫丰源

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1. 交叉熵损失定义与化简

交叉熵损失定义为

$$J(\boldsymbol{y}, \hat{\boldsymbol{y}}) = -\sum_{w \in \text{Vocab}} y_w \log \hat{y}_w.$$

因为 $\boldsymbol{y}$ 是 one-hot vector，

$$\text{当} w = o \text{ 时} y_o = 1, \qquad \text{当} w \neq o \text{ 时} y_w = 0.$$

代入后只有 $w = o$ 的那一项不为 0：

$$J(\boldsymbol{y}, \hat{\boldsymbol{y}}) = -y_o \log \hat{y}_o - \sum_{w \neq o} 0 \cdot \log \hat{y}_w = -y_o \log \hat{y}_o.$$

又因为 $y_o = 1$，所以

$$J(\boldsymbol{y}, \hat{\boldsymbol{y}}) = -\log \hat{y}_o.$$

2. 求 $\dfrac{\partial J}{\partial \boldsymbol{v}_c}$

损失

$$J(\boldsymbol{y}, \hat{\boldsymbol{y}}) = -\log \hat{y}_o.$$

先对 $\hat{y}_o$ 求导：

$$\frac{\partial \hat{y}_o}{\partial \boldsymbol{v}_c} = \hat{y}_o \left(\boldsymbol{u}_o - \sum_w \hat{y}_w \boldsymbol{u}_w \right).$$

由 chain rule：

$$\frac{\partial J}{\partial \boldsymbol{v}_c} = -\frac{1}{\hat{y}_o} \cdot \frac{\partial \hat{y}_o}{\partial \boldsymbol{v}_c}.$$

代入化简：

$$\frac{\partial J}{\partial \boldsymbol{v}_c} = \sum_{w \in \text{Vocab}} (\hat{y}_w - y_w) \boldsymbol{u}_w.$$

因为 $\boldsymbol{U}$ 的每一列正是 $\boldsymbol{u}_w$，上式是矩阵-向量乘法：

$$\sum_w (\hat{y}_w - y_w)\boldsymbol{u}_w = \boldsymbol{U}(\hat{\boldsymbol{y}} - \boldsymbol{y}).$$

因此

$$\frac{\partial J}{\partial \boldsymbol{v}_c} = \boldsymbol{U}(\hat{\boldsymbol{y}} - \boldsymbol{y}).$$

3. 求 $\dfrac{\partial J}{\partial \boldsymbol{u}_w}$

因为 w 决定了 y_w 的取值，分 $w = o$ 或 $w \neq o$ 两种情况。

损失

$$J = -\log \hat{y}_o, \qquad \hat{y}_o = \frac{\exp(\boldsymbol{u}_o^\top \boldsymbol{v}_c)}{Z}, \qquad Z = \sum_k \exp(\boldsymbol{u}_k^\top \boldsymbol{v}_c).$$

当 $w = o$：

$$\frac{\partial J}{\partial \boldsymbol{u}_o} = (\hat{y}_o - 1)\boldsymbol{v}_c = (\hat{y}_o - y_o)\boldsymbol{v}_c.$$

当 $w \neq o$，$\boldsymbol{u}_w$ 只在分母 Z 中：

$$\frac{\partial J}{\partial \boldsymbol{u}_w} = \hat{y}_w \boldsymbol{v}_c = (\hat{y}_w - y_w)\boldsymbol{v}_c,$$

因为此时 $y_w = 0$。

综上：

$$\text{当} w = o, \quad \frac{\partial J}{\partial \boldsymbol{u}_o} = (\hat{y}_o - y_o)\boldsymbol{v}_c;$$
$$\text{当} w \neq o, \quad \frac{\partial J}{\partial \boldsymbol{u}_w} = (\hat{y}_w - y_w)\boldsymbol{v}_c.$$